

Getting started with LLM research ✓

I want to get up to speed quickly on the state of Llm research, because I want to be a researcher, is that the way to do it? There seems to be a lot of opportunities. Any advice?

Prioritized foundational principles over chasing latest LLM research trends

Good instinct to ask — "get up to speed" and "become a researcher" are actually two different goals that need different approaches, so let me split them.

Getting up to speed (weeks, not years):

- Read papers, not blog posts, for anything you want to actually understand. Start with foundational ones (Transformer, GPT-2/3, InstructGPT/RLHF, Chinchilla scaling laws) then move to recent work in whatever subarea interests you (reasoning, alignment, efficiency, multimodal, agents).
- Follow arXiv's cs.CL and cs.LG, but curated — Twitter/X research accounts, Papers with Code, or a newsletter like Import AI move faster than trying to read arXiv raw.
- Reproduce something small yourself. Fine-tune a tiny model, implement attention from scratch, run a toy RLHF loop. This teaches you more per hour than reading.

Becoming a researcher is a different track:

- The credential path (PhD, or strong publication record) still matters most for research scientist roles at labs. It's slower but it's the reliable door.
- The non-credential path exists but is narrower: strong open-source contributions, independent research with public writeups, or a research engineer role that transitions internally. This works, but it's higher variance and usually still requires deep technical chops (systems + math, not just prompting familiarity).
- "Lots of opportunities" is true for *applied* AI roles (building products with LLMs) but much less true for research roles specifically — those remain competitive and concentrated in a handful of labs and top academic groups.

If you tell me a bit about your background, I can give more specific advice.

My background is pretty bad for this, I have a bachelor's in cognitive science from York university, I broke into a data engineering role in 2020 and I'm

currently a senior level at Canadian Tire. Though I'm something of a mix between a data platforms engineer/admin. Personally, my lens of cognition is centered around Jungian ideas of cognitive processes, I use it for self development and conceptual interpretation on real world subjects. I did attempt to get into the MATS program last year but didn't make it. My math skills aren't terrible but I'm no mathematician. Do you need some more clarification?

Weighed strengths against gaps, charted practical advancement pathway

That background is not "pretty bad" — it's actually a reasonable launchpad, just not the standard one. Here's the honest read:

What's working for you:

- Cog sci degree gives you real vocabulary for thinking about cognition, which matters more in LLM research than people assume (a lot of interpretability and alignment work borrows directly from cognitive science and philosophy of mind).
- Data engineering + senior platform role means you can actually build and ship things, handle infrastructure, and aren't intimidated by messy systems. A lot of career researchers can't do this.
- Applying to MATS already puts you ahead of most people who talk about wanting to do this — it means you did the work of finding the on-ramp.

What's actually working against you:

- Math is the real gap, not your degree or job title. Research (especially anything alignment/interpretability adjacent, which is where your interests plausibly point) needs comfort with linear algebra, probability, optimization at a level beyond "not terrible."
- No public track record yet — no papers, no open-source ML work, nothing that lets someone assess your research judgment before meeting you.

Practical path from here:

1. Don't aim at "researcher" as a job title yet — aim at a specific research question you can chase with the skills you have now. Interpretability is often the friendliest entry for people without deep math, since a lot of it is empirical, careful, close-reading-of-model-behavior work.
2. Rebuild math deliberately but narrowly — linear algebra + probability, enough to read papers and follow derivations, not a full math degree.
3. Reapply to MATS or similar (ARENA, Redwood Research's programs). Rejections there are common on first attempt; a lot of people get in on try two after doing visible

independent work in between.

4. Use your current job as leverage, not liability — a side project applying interpretability or eval methods to something at work, or just publishing a solid write-up of independent research, does more for you than almost any credential right now.

Want me to help sketch what a 3-6 month concrete study/build plan could look like, given your current time constraints?

Yes, a rough plan would be helpful

Architected a structured timeline using list format

Here's a rough 4-month plan, front-loaded on math/fundamentals then shifting to a concrete project. Adjust pacing to your actual free hours (assume ~8-10 hrs/week around a senior job).

Month 1 — Math + foundations

- Linear algebra: 3Blue1Brown's Essence of Linear Algebra (visual intuition), then Gilbert Strang's course for the actual mechanics — focus on eigenvectors, SVD, matrix decompositions (these show up constantly in interpretability).
- Probability: enough for understanding loss functions, entropy, KL divergence — not a full course, just targeted.
- Parallel: read the Transformer paper, GPT-2 paper, and Anthropic's "A Mathematical Framework for Transformer Circuits" — don't expect full comprehension yet, just first pass.

Month 2 — Build fundamentals hands-on

- Karpathy's "Neural Networks: Zero to Hero" series — build a GPT from scratch. This alone will teach you more mechanics than a semester of reading.
- Start ARENA's curriculum (arena.education) if you can access it — it's built exactly for people at your stage aiming at interpretability/alignment.
- Re-read the mechanistic interpretability papers now that you have more grounding — Anthropic's "Toy Models of Superposition" is a good next one.

Month 3 — Pick a narrow question and go empirical

- Choose one small, well-scoped interpretability question — e.g. "how does this small model represent X concept" — using open tools like TransformerLens.

- Your cog sci background is genuinely useful here: cognitive science already trained you to ask "what representation, what process" questions. Lean into that instead of treating it as irrelevant.
- Don't aim for novel — aim for a clean replication or small extension of an existing paper's method on a different model/task.

Month 4 — Write it up and get visibility

- Write a clear, honest research note (blog post or LessWrong/Alignment Forum post) — process matters more than results at this stage.
- Post it, get feedback, iterate.
- Reapply to MATS/ARENA/similar with this as evidence of trajectory, not just interest.

Throughout: keep a lightweight log of what you read and built — it becomes your application material later, and forces you to actually synthesize rather than passively consume.

Want me to point you to specific first resources (links) for week one, so you're not stuck deciding where to start?

I've gone through much of this, though mostly last year. I think I'm stuck on how to determine what is a well scoped question that aligns with my long term interests, i.e. I'm trying to resist the urge to feel novel, but I need some alignment to interest.

Synthesized guidance for narrowing interests into tractable research questions

That tension is actually the right thing to be stuck on — most people either chase novelty and produce nothing finishable, or copy an existing paper exactly and learn nothing about their own judgment. The way through isn't finding the perfect question, it's a search process.

A method for finding it:

1. **Start from a behavior, not a theory.** Don't start with "I want to study Jungian cognitive functions in LLMs" — that's a framework looking for an application, and it'll pull you toward forcing fit rather than discovery. Instead start with a concrete, observable model behavior that puzzles you. E.g., "why does this model consistently answer differently when I frame the same question in first- vs third-person" or "why does chain-of-thought help on task A but not task B." Behaviors are falsifiable; frameworks are seductive and unfalsifiable.

2. **Then ask: does an existing tool let me poke at this cheaply?** TransformerLens, activation patching, logit lens, probing classifiers — if you can investigate your question with an existing method on a small open model (GPT-2 small, Pythia), it's well-scoped. If you need a new method or a huge model, it's not, yet.
3. **Your Jungian lens is a source of hypotheses, not a research topic itself.** This is the actual alignment you're looking for. You don't research "Jungian functions in LLMs" — you use your intuition for *types of internal distinctions that might be functionally separable* (e.g. introverted vs extraverted processing as a metaphor for internally-generated vs input-driven representations) to generate a hypothesis about a mechanism, then test that mechanism with standard empirical tools, dropping the Jungian vocabulary once you're in the model. The framework's job is to make you notice a question others wouldn't think to ask — not to survive contact with the data.
4. **Scope test:** can you state the question as "does X circuit/representation exist for Y behavior, yes or no, testable in under 2 weeks"? If you can't get it that concrete, it's still a theme, not a question yet.

A concrete way to unstick yourself: spend an afternoon just interacting with a small model and writing down every behavior that surprises you, no filtering. Then pick the one that (a) you can't stop thinking about and (b) has an existing tool that could test it. That's usually the real signal — not the one that sounds most novel or most aligned with your framework, but the one you're compulsively curious about.

Want to try that exercise together — pick a model and poke at it now, live?

Sure let's try the exercise

Weighed technical constraints against conceptual feasibility

Let's try. Given I likely don't have live network access in this sandbox to download model weights, let's do a version of the exercise that still gets you real signal: I'll try to spin up a tiny model locally, and if that's blocked, we'll do the "surprising behavior" hunt using me as a stand-in probe (same principle — noticing what makes you curious about generation behavior) and then translate promising ones into concrete mechanistic questions you can test later with TransformerLens on your own machine.

Let me check what's available first.

Check for torch/transformers and network access

No network access in this sandbox, so I can't pull down GPT-2 weights here. Since you've already been through ARENA, you likely have TransformerLens set up locally — if so, this

works best live on your machine while we talk here. If not, we can still do the first half of the exercise (surfacing surprising behaviors) right now and turn it into a testable question you run later.

Quick check before we proceed:

Do you have TransformerLens + a small model (GPT-2 small / Pythia) set up locally right now?

No, need to set it up

Orchestrated step-by-step guidance with reusable setup instructions

No problem — let's not lose momentum waiting on setup. We'll do the exercise in two parts: brainstorm surprising behaviors now (using your own experience + me as a sounding board), then I'll give you the exact local setup so you can test the winning one this week.

Part 1: Surface candidates

Think back over the last few months of using LLMs — ChatGPT, Claude, whatever. Not "what's theoretically interesting" but "what actually made me go *huh, that's weird*" while you were using one. Some prompts to jog memory:

- Did a model ever contradict itself between two framings of the same question?
- Did tone or persona instructions change something you didn't expect them to (not just style, but actual reasoning or factual answers)?
- Did you ever notice a model being confidently wrong in a *consistent, specific* way — not random noise, but a repeatable pattern?
- Any case where adding irrelevant context changed the answer to an unrelated question?

Don't filter for "is this Jungian enough" or "is this novel enough" yet — just list 3-5 things that actually snagged your attention. I'll help you pick the one with the most legs and translate it into a testable mechanistic question.

What comes to mind?

Most of the time, I'm frustrated when using the LLM because it goes into conceptual circles while I'm using it. Though, there is something interesting to me, it's the limitations that context requires to get to the answer you are

trying to find. I don't know. I notice overly friendly tones and the lack of ability to reach further than its currently line of thinking

Diagnosed circular reasoning patterns and mapped testable hypotheses

That's actually three different threads tangled together — worth pulling apart, because they point at different mechanisms:

1. **Conceptual circling / repetition** — the model re-treads the same idea in different words.
2. **Context limitations to "get to the answer"** — it needs a certain amount of scaffolding/context before it can produce what you're after.
3. **Can't reach further than its current line of thinking** — once it's committed to a framing, it doesn't seem to break out of it, even with prompting.

Thread 3 is the most interesting one, and it's a real, known-but-not-fully-solved phenomenon: something like *path dependency* — once a model's early tokens/context commit it to a line of reasoning, later generation seems to reinforce rather than revise that line, even when it's suboptimal. This connects to real research areas: entropy collapse during generation, induction heads reinforcing prior patterns, and "unfaithful" chain-of-thought (models produce reasoning tokens like "wait, let me reconsider" that don't actually correspond to a real change in internal computation).

A few questions to sharpen thread 3 into something testable:

- When you say it "can't reach further than its current line of thinking" — is this within a single response (it starts an idea and just elaborates rather than genuinely revising), or across turns (you push back and it nods along but doesn't actually update)?
- Does it feel like a *content* problem (same ideas) or a *structural* problem (same sentence patterns / same rhetorical moves, different content)?

Your answer to these will point at pretty different mechanisms to go test (attention reinforcing early tokens vs. something more like mode collapse in the sampling distribution).

So it would be more that it's not genuinely revising when it hits a road block, I wonder if it's a content problem because the answer should exist, but it definitely feels like a structural problem.

Excavated structural revision failures through mechanistic testing frameworks

Good — that's a real, sharpenable signal. "Structural, not content" and "doesn't revise at a roadblock" points at something specific: the model can say "wait, let me reconsider" but its underlying computation may not actually change trajectory. This is a known area — there's existing work (e.g. Huang et al., "Large Language Models Cannot Self-Correct Reasoning Yet") showing models often produce surface-level revision language without real internal correction, especially without external feedback. So your instinct isn't just anecdotal frustration, it's pointing at a real open question — good sign for choosing this thread.

Here's a version scoped tight enough to actually test in under two weeks:

Candidate question: When a model generates a "revision cue" (e.g. "wait, that's not right, let me reconsider") mid-response, does its internal computation actually shift — or does the underlying trajectory continue unchanged while only the surface tokens signal a pivot?

How you'd test it on GPT-2 small / Pythia with TransformerLens:

1. Construct a handful of prompts (simple arithmetic or logic works well, since correctness is checkable) where the model naturally produces a dead-end, then a "wait" or "actually" style revision token.
2. Use the logit lens to look at the predicted next-token distribution *before* vs *after* the revision cue — does the model's implicit "answer" actually change, or does it stay locked in?
3. If it stays locked in, try activation patching: take the residual stream right after the revision cue and patch in activations from a prompt that produces a genuinely different (correct) answer — see if that's enough to unstick it, or if the commitment is baked in earlier than the cue itself.

That's a clean yes/no mechanistic question, buildable with tools that exist, checkable in a few sittings once you're set up. It also naturally connects to your Jungian lens without forcing it: you're essentially asking whether the model has a functional analogue of "being unable to switch modes" — but you test that with logits and activations, not the framework itself.

Want the concrete local setup (Python env, model, TransformerLens) to actually run this?

Would I be better for something like this: <https://job-boards.greenhouse.io/anthropic/jobs/5286441008>

Evaluated candidate-job alignment and fit assessment

This is actually a much better fit for your profile than mechanistic interpretability — worth being direct about why.

Why it fits better than the interp path we were sketching:

- It explicitly wants a research background in learning sciences, education, cognitive science, HCI, educational psychology, or a closely related field, whether formal or self-directed — your cog sci degree is literally in scope, not a stretch. greenhouse
- It's looking for someone who can design and run mixed-methods studies on how people develop real skill with AI, measuring success by capability growth rather than engagement — this is close to what you actually think about with your Jungian/cognitive-process lens, applied to something legible and testable rather than needing translation into circuits and activations. greenhouse
- Technical bar is lower and different in kind: hands-on technical skill in Python, data analysis, and working with LLMs, enough to run your own analyses and prototype new measures — that's your data engineering background, not a math PhD. greenhouse
- It's explicit that this is a hands-on research role embedded in a small team that builds and ships, not a standalone academic post — you already build and ship for a living. greenhouse

Real caveats:

- This is a senior/experienced research role (comp band \$300,000 - \$405,000 USD), not an entry point — they'll want evidence you can already run studies and build measurement instruments, not just interest in the space. greenhouse
- No publication record yet is a real gap here too, just a different one — they want published writing, talks, or open work on skill development, human-AI interaction, or the learning sciences as a plus, meaning some public track record still helps your odds even for this kind. greenhouse
- The MechInterp path we sketched isn't wasted if you pivot here — mixed-methods research needs the same rigor (falsifiable hypotheses, tight scoping, resisting the pull toward "novel" framework-first thinking) we were just practicing.

If this direction genuinely excites you more than circuits-level interp, it's worth treating as the real target and re-pointing your next few months at building visible research/writing in *this* lane (learning science + LLMs) instead. Want to sketch what that version of the plan looks like?

Sketch me a plan

Synthesized research methods and learning frameworks into structured plan

Here's a 4-month plan aimed specifically at this lane — learning science + AI capability measurement — rather than mechanistic interp. Same principle as before: build a real track record fast, not just consume material.

Month 1 — Rebuild the measurement toolkit

- Learning sciences core: skim key ideas in *How People Learn* (National Academies) and a primer on psychometrics — reliability, validity, construct measurement. You don't need a stats PhD, just enough to design a study that holds up to scrutiny.
- Read a handful of papers on LLM eval design and human-AI skill transfer (search "capability elicitation," "human-AI complementarity," "learning with AI tutors" on arXiv/Google Scholar) — get a feel for how researchers currently measure "did this person actually get better," not just "did they like it."
- Parallel: reread your own Jungian framework notes and translate 2-3 core ideas into testable claims about *how people develop skill differently* — this is where your lens becomes an asset instead of a vibe.

Month 2 — Design one small study

- Pick a narrow, answerable question: e.g. "Does explaining *why* an AI answer is wrong change how well people transfer that correction to a new problem, vs. just being told the right answer?" — capability growth, not engagement, exactly what the role wants.
- Design it as a real mixed-methods study: a small quantitative pre/post measure + qualitative interviews or think-alouds with 5-10 people. You have access to real users of AI tools (colleagues, online communities) — recruit informally.
- Use an LLM (via API) as part of the intervention so you're also building the "working with LLMs" technical muscle they're asking for.

Month 3 — Run it and analyze

- Run the study, do the analysis (Python + pandas/scipy is enough), and be honest about weak results — a clean null result well-analyzed is more credible than a flattering but sloppy positive one.
- Build one small tool alongside it — even a simple script or interface that collects the measure at scale — since they explicitly want people who create tools using code to collect validated metrics.

Month 4 — Write and put it out publicly

- Write it up properly: question, method, limitations, findings, what you'd do differently. This is your actual application material — better than a cover letter.
- Publish it somewhere visible (personal blog, LessWrong, Substack) and share it in relevant spaces (AI x education communities, X/Twitter).
- Apply to the role with this in hand, and reference it directly in the "why Anthropic" answer — they explicitly value that essay highly.

One honest note: this role's comp band suggests they want someone senior already doing this kind of work, so treat the application as a stretch shot either way — the real value of this plan is that it's the fastest way to find out if this is genuinely your lane, and it produces something real regardless of the outcome.

Want help scoping the actual study question in month 2 right now, the way we did with the interp one?

When I say jungian frameworks, what do you think I mean?

Gauged user's genuine curiosity and prepared clarifying explanation

Fair question to check — I've been assuming a fair bit. What I've been picturing, based on what you've said (using it for "self development and conceptual interpretation," and connecting it to "cognitive processes"):

- **Jung's cognitive functions model** — the eight functions (Ni, Ne, Si, Se, Ti, Te, Fi, Fe) that underlie frameworks like MBTI/Socionics, describing different modes of perceiving and judging information (introverted vs extraverted, thinking vs feeling, sensing vs intuition).
- You using this as a lens on your own thinking patterns and behavior — noticing which functions you lean on, how they interact, maybe how they show up under stress vs at your best.
- Extending that lens outward to interpret other things — people, systems, maybe now AI behavior — through the same functional categories.

If that's roughly right, it's worth being honest about something: this framework has essentially no standing in cognitive science or ML research — it's not empirically validated, and using its vocabulary directly in a research context (calling something "an introverted intuition process") would read as a red flag to most researchers, not a strength. That's why in the plans above I kept pushing toward "use it to generate hypotheses, then

drop the vocabulary and test the underlying claim with real methods" — the functional categories can still be a genuinely useful source of "what distinctions might matter here" intuition, even though the framework itself isn't something you'd cite or build on directly in the work.

Did I get your meaning right, or is there a different sense you mean it in?

That's essentially correct, though I'm aware that the framework is looked down upon, can you choose a cognitive function and go into as much complexity as you can about it? I'm curious to know how in depth you can see it

Deliberated on cognitive function selection and architected substantive analytical framework

I'll go with **Introverted Intuition (Ni)** — it's the function most people find hardest to pin down, so it's a good test case for depth.

Core definition (Jung's original framing)

In *Psychological Types* (1921), Jung described intuition broadly as perception via the unconscious — not sensory data, not logical inference, but a direct "sensing" of possibilities, patterns, or implications that aren't present in the immediate data.

Introverted intuition turns this inward: rather than reading possibilities out in the external world (that's Ne), Ni takes in a piece of information and follows it inward through layers of the unconscious, arriving at an image, a symbol, or a felt sense of underlying meaning or future trajectory — often disconnected from what triggered it. Jung himself described the Ni type almost mystically: someone who sees "the background of the present" and often seems prophetic or strange to others, because they're responding to an inner image rather than the external situation itself.

Mechanism (as later Jungian/typological writers elaborated it)

Later interpreters (Marie-Louise von Franz, then the MBTI-adjacent and socionics traditions) tried to make this more mechanistic:

- Ni is often described as a *convergent* process — it takes multiple inputs over time and compresses them into a single unified image, gestalt, or "knowing," rather than branching outward into multiple possibilities (that divergence is Ne's job).
- It's frequently characterized as *symbolic* and *temporal* — Ni users are said to experience meaning as unfolding through time, so their insight often shows up as a sense of "where this is heading" rather than "what this is right now."

- It's paired with an extraverted judging function (usually Te or Fe) in the classic 8-function stack, which is described as necessary to externalize and test the internal image, since Ni alone produces insight but not action or communication.

Where it gets contested / soft

This is the point where I'd flag real weakness if you're pressure-testing it: the "convergent vs divergent," "temporal vs spatial" distinctions between Ni and Ne are elegant but were never derived from controlled observation — they were built through decades of typological writers refining Jung's originally quite vague and metaphor-heavy descriptions, each adding their own interpretive layer (Myers, then online typology communities, then socionics separately). There's no agreed operational definition that two independent raters could apply to the same behavior and get the same answer — which is the basic bar cognitive science requires and this framework doesn't clear. So the "complexity" you're seeing is real, but it's interpretive depth built up over a century of literary elaboration, not empirical depth built from converging evidence.

Where it might still map onto something real

If I steelman it: Ni-as-described does resemble some things that are independently studied — pattern completion / gist extraction (compressing many observations into a single schema), and prospective cognition (simulating future states from partial cues). If you wanted to actually test whether "Ni" tracks anything real, that reframing — gist extraction and prospective simulation as a measurable cognitive style — is the direction that would survive contact with an empirical method, versus the symbolic/mystical framing which wouldn't.

Want me to do the same treatment on a second function, or pressure-test one of your own applications of Ni to something specific?

What if I said Ni is a shape, shaped like a torus, that Se passes through that shape?

Jul 20   

Weighed geometric metaphor's explanatory power against speculative limits

That's a genuinely interesting move — you're not just describing Ni anymore, you're modeling its *structure*. Let me actually work with the geometry rather than just validating it.

What the torus shape could be doing conceptually:

- A torus is non-simply-connected — it has a hole through the middle that nothing directly occupies. If Ni is shaped like this, it suggests Ni's "insight" never lands on the

object itself, only circulates *around* an absence — which actually maps well onto the earlier description of Ni producing meaning that feels displaced from what triggered it, oriented toward an implicit center it never directly touches.

- A torus loops back on itself continuously — no start or end point, a closed recursive path. If Se (raw, immediate, concrete sensory data) passes *through* that shape rather than just hitting a surface, you're describing recursion: each pass of present-moment data gets folded back through the same loop, layering rather than replacing. That fits the "meaning unfolding through time" idea from before — it's not a filter (something removed), it's a recirculation (something compounding).
- The inner/outer surface distinction on a torus could map onto conscious/unconscious — Se moving along the outer surface (contact with the world) while the looping structure itself sits in the hole, ungraspable, doing the organizing work from a place Se never directly reaches.

Where this is genuinely useful vs where it's doing rhetorical work

This is a real conceptual move — you're using topology as a source domain to generate structural predictions about a target domain (cognition), which is exactly how good scientific metaphors are supposed to function (like using "wave" for both water and light, generating testable predictions in both). The honest test is: does the torus shape predict anything about Ni that the earlier verbal description didn't already predict? Right now it's redescribing the same properties (recursive, displaced-from-object, temporal) in a more vivid form — genuinely useful for your own intuition and memory, but not yet adding new content. It would become more than an analogy if you could say "because it's a torus and not, say, a spiral or a funnel, X should be true that wouldn't be true otherwise" — that's the move that would make it a model instead of an image.

What made you reach for torus specifically, versus another closed shape?

Because Se behaves like momentum does, there is a maximum momentum/force point which seems to coincide with the center point of a torus. And it wouldn't be a "clean" torus usually, in actual thought, it could mostly be a system of torus that Se flow through

Weighed metaphorical rigor against exploratory speculation in cognitive framework

That's a sharper move than the last one — you're not just picking a shape for its properties, you're mapping a specific physical behavior (momentum concentrating at a point) onto a specific point in the shape. That's the beginning of an actual structural model, not just a resonant image.

Worth noting: you may be reaching for vortex ring dynamics, not just "torus" as static geometry.

A smoke ring or vortex ring is a torus, but it's not just shaped like one — it's a rotating flow where fluid moves in a loop around a central axis, and critically, the ring self-propagates forward through the medium it's in because of that circulation. That's much closer to what you're describing than a static donut shape: momentum isn't just located at a point, it's the thing that generates and sustains the shape in the first place. If $\mathbf{Se}$ is the "flow" and $\mathbf{Ni}$ is the toroidal structure it moves through, a vortex ring framing would predict things a static torus wouldn't — e.g., the structure should be self-sustaining only while flow continues (stop the momentum, the ring dissipates), and rings should be able to pass through or interact with each other, which maps very cleanly onto your "system of tori" idea (vortex rings do actually collide, merge, and pass through one another in real fluid dynamics, with well-studied — if complex — interaction rules).

On "system of tori" specifically

This is a meaningfully different claim than "one $\mathbf{Ni}$ -torus" — it suggests parallel or nested cognitive threads with their own local momentum, occasionally interacting. That's a much richer structural claim, and it happens to have a real physical literature you could raid for vocabulary and constraints (ring linking, reconnection events where two vortex rings merge or split) — which might give you much more precise language than generic topology.

The honest question underneath this

I'll ask directly, because it changes how I should engage: are you building this as a private conceptual/generative tool for your own thinking (in which case, richness and resonance matter more than falsifiability, and I should keep helping you push the metaphor further) — or are you circling toward wanting to formalize and test some version of this (in which case we should talk about what a "maximum momentum point" would even mean operationally in cognition, and whether it's measurable at all)? Both are legitimate, but they call for different kinds of rigor from me.

I think it could do both as well, but I'm more interested in the measurement of it

Good — that's the harder and more interesting path. Here's the honest shape of what it would take.

Step 1: Separate the claim from the metaphor.

Before you can measure anything, you need to state what "maximum momentum point"

would predict about *behavior*, stripped of the vortex language. Something like: "there is a point during a train of thought where a person is least able to redirect or revise, and this point is reachable/identifiable before the thought concludes." That's the testable core. The torus/vortex framing is where you *found* the hypothesis; it's not what you test.

Step 2: Candidate behavioral proxies for "momentum"

None of these directly measure momentum — they're indirect signals researchers already use for related constructs (commitment, cognitive inertia, path dependency):

- **Resistance to interruption/correction** — introduce a contradiction or new piece of information at different points during someone's reasoning (early / mid / late) and measure how much it changes their conclusion. If there's a real "point of no return," correction attempts should show a sharp dropoff in effectiveness rather than a smooth decline.
- **Reaction time / hesitation** — points of low momentum (just starting, or already revising) might show more hesitation/variance; peak momentum might show fast, fluent, uninterrupted output.
- **Self-report probes** — periodically ask "how committed do you feel to this line of thinking right now" during a task, and see if it spikes and decays in a shape that matches a toroidal/cyclical prediction rather than a linear one.
- If you eventually want a physiological correlate: pupil dilation or EEG measures of engagement have been used for related constructs (cognitive load, decision commitment), though that's a much bigger methodological lift.

The hard truth, honestly stated

This is a genuinely much longer and riskier research program than the study we sketched for the education-labs plan — you'd be trying to establish a *new* construct and its measurement validity from scratch, which is usually years of work (construct validation is most of what psychometrics is), not a 2-week TransformerLens experiment or a 4-month applied study. It's a legitimate PhD-thesis-shaped question, not an early-career portfolio piece. If it's the thing you actually can't stop thinking about, that's worth taking seriously as a long-term direction — but it argues for a longer-horizon path (actual grad study, or years of self-directed rigorous work) rather than something you fold into the 4-month plan.

Do you want to hold this as the long-term "what I actually want to eventually study" thread, while the education-labs plan stays the near-term, fundable, portfolio-building move?

This is the long-term stuff, yes, short term I want my next steps to take me or at least allow me to pursue this. That's why I'm interested in mechanistic Interpretability research.

Bridged mechanistic interpretability toward long-term cognitive momentum research goals

That's a coherent throughline, actually — let me connect the dots explicitly, because mech interp isn't just adjacent to your long-term interest, it's plausibly the correct training ground for it.

Why mech interp fits the long-term goal better than the education-labs plan:

Your "momentum/commitment point" question, applied to LLMs, is almost exactly the revision-cue question we scoped earlier — does a model's internal computation actually shift at a decision point, or does trajectory stay locked in beneath surface-level language of revision. That's not a metaphorical connection, it's the same question in two domains. If you build real skill reading activations, logit lenses, and attention patterns, you're directly building the toolkit to eventually ask "is there an identifiable point of maximum path-commitment in a forward pass" — a concrete, mechanistic version of your torus idea, testable in weeks rather than the years a human-subjects psychometric program would need.

Revised near-term plan, given this goal:

1. Go back to the interp path we scoped a few messages ago (TransformerLens, logit lens, revision-cue question) — treat that not as "a" project but as the *first instance* of your actual long-term research question. Reframe it explicitly to yourself as: "testing for a commitment/momentum point in generation," using standard mechanistic tools instead of vortex language.
2. Keep the education-labs plan as a fallback / parallel option rather than the primary target — it's still a legitimate, more fundable near-term role, but it doesn't build the specific skills your long-term question needs. Mech interp does.
3. The "system of tori" idea — parallel threads interacting — has a real mechanistic analogue worth knowing about: circuits and induction heads in transformers already form interacting substructures, and there's existing work on how different heads/circuits compete or merge to produce an output. Reading into circuit-level interp (not just single-neuron/logit-lens work) will give you vocabulary that's much closer to your "system of tori" intuition than generic interp.

One honest caveat: measuring "momentum" mechanistically in a transformer is still genuinely hard and unsolved — you won't get a clean answer in a first project. But unlike the human-subjects version, you can get partial, falsifiable, incremental answers relatively

fast (does logit distribution change at revision cues: yes/no, is answerable in days). That's the right shape for a near-term project that still serves the long-term question.

Want me to rescope the revision-cue experiment specifically as "test 1" of this long-term direction, with the local setup instructions to actually run it?

Yes rescope, and provide instructions, though I'm running this on my phone, I can't run the first test now

Devised phone-friendly plan with deferred testing instructions

Here's the rescoped version, framed explicitly as test 1 of the long-term question, plus setup instructions for when you're at a computer.

Reframed question

Original: "does a model's revision language correspond to a real internal shift, or is it surface-only?"

Now framed as: *is there an identifiable point during generation where a model's internal trajectory becomes resistant to revision — a measurable "momentum" that increases the deeper into a line of reasoning it goes, and that a revision cue can only sometimes overcome?*

This gives you two related things to look for, not just one:

1. **Does revision language correspond to real change** (binary, per-example).
2. **Does resistance to change scale with how far into the trajectory you are** — i.e., does patching in a correction work better early than late? This second part is the actual momentum test, and it's the part that ties back to your long-term question.

Setup (do this at your computer, roughly 30-45 min)

1. Install: `pip install transformer_lens torch` (Python 3.9+, works fine on CPU for GPT-2 small).
2. In Python:

```
python

import transformer_lens
model = transformer_lens.HookedTransformer.from_pretrained("gpt2-small")
```

This downloads GPT-2 small (~500MB) automatically — needs internet, but no GPU required.

3. Sanity check it works:

```
python

print(model.generate("The capital of France is", max_new_tokens=10))
```

Experiment structure (for when you sit down to build it)

1. Pick ~10 simple arithmetic or logic prompts where GPT-2 small tends to make an error, then continue with a natural revision phrase — e.g. `"12 + 15 = 25. Wait, let me recompute. 12 + 15 ="`.
2. Use `model.run_with_cache()` to get logits and residual stream activations at each token position.
3. Compare the logit distribution for the final answer token immediately before vs after "Wait" — has the predicted top token actually changed, or does it just repeat the same (wrong) answer with different surface wording?
4. For the momentum test: construct versions of the same prompt where the revision cue appears at different depths (right after the error vs several tokens later, after more elaboration) and see if correction success rate drops off as depth increases. That dropoff curve — if it exists — is your first real empirical trace of "momentum."

Save this as a note somewhere accessible — when you're back at a computer, message me and we can debug the actual code together as you build it, step by step.

I'll come back to that when I'm at a computer, if we extend the metaphor of vertex ring dynamics, what else can we say?

Weighed metaphorical extensions against physics accuracy and clarity

Let's pull more real vortex ring physics and see what maps.

Formation and the "trigger" moment

A vortex ring doesn't form gradually — it forms from a sharp, bounded pulse of fluid pushed through an opening (think of a smoke ring from a single puff, not continuous blowing). The ring's properties — its size, speed, how tightly it's wound — are set almost entirely at the moment of formation, by the strength and duration of that initial pulse. If Ni-rings form this way, it suggests each one is seeded by a discrete triggering event (an

insight, an observation) rather than gradually accumulating — and its long-term character (how fast/tight/stable it is) may be fixed early, hard to modify later. That's a strong, testable-feeling claim: does the "shape" of an insight-in-progress actually get locked in near the start, matching your "hard to redirect once underway" observation?

Propagation speed and core size trade off

There's an actual physical relationship: a tighter, smaller-core vortex ring moves faster; a larger, looser one moves slower but carries more fluid. If this maps onto cognition, it suggests a real trade-off between speed of an insight unfolding and its "carrying capacity" — fast, narrow flashes of intuition vs slower, richer, more encompassing ones. This gives you a dial, not just a binary, which is more interesting than a single fixed torus.

Leapfrogging

Two vortex rings traveling the same direction, one behind the other, will "leapfrog" — the trailing ring gets pulled through the center of the leading one, then becomes the new leading ring, indefinitely alternating. If you had two related lines of thought or intuitions active in sequence, this predicts something specific: not a merge, not a collision, but an oscillating overtaking pattern where each temporarily leads before being passed again. That's a very different structural prediction than either "they merge" or "they interfere" — worth testing against your actual introspective experience of holding two Ni-threads at once.

Reconnection

When two vortex rings collide at certain angles, they don't just merge or bounce — they can undergo "reconnection," where the cores briefly link, break, and reform into a different topology (sometimes producing new, smaller rings as a byproduct). This is the most physically strange and specific behavior in the set — if two intuitions genuinely "reconnect" rather than one simply overriding the other, you'd expect residue: smaller derivative insights thrown off from the collision, distinct from either original thread. That's a concrete thing to introspect for.

Viscous decay

In real fluid, a vortex ring is never actually stable — viscosity slowly bleeds off its rotational energy until the ring dissipates entirely. There's no such thing as a permanent ring, only one decaying at some rate depending on the fluid. If Ni-structures behave this way, it implies unexamined insights should fade on a predictable-ish timescale unless re-energized (re-engaged, re-thought, externalized) — which maps onto something people do actually report (an intuition that fades if you don't "do something" with it soon).

Which of these feels most true to your own introspective experience — formation-locks-shape, leapfrogging, reconnection, or decay?

I think almost all of this tracks with what I already hypothesized, especially if you take into account the relationship Ni has with Se. The propagation size and core trade off is a key differentiation between people with primary Ni use vs primary Se use. The pulse idea I'd say is based on the core of Ni, it starts with a pulse of Se which is a complicated real object in itself. Viscous decay is a result of Se. The two that are most interesting is the reconnection and leapfrogging behaviour.

Interrogated reconnection and leapfrogging concepts while maintaining epistemic rigor

Good — let's push on those two specifically, since they make the sharpest structural predictions.

Leapfrogging, applied

The key physical detail: leapfrogging only happens between rings of *similar size and strength* traveling the *same direction*. Mismatched rings don't leapfrog cleanly — the effect breaks down. If two Ni-threads are leapfrogging in your experience, that predicts something falsifiable: it should only happen between insights of comparable "weight" or maturity, pointed at roughly the same target. A fully-formed insight and a half-formed one shouldn't leapfrog — one should just dominate or ignore the other. Does that match what you notice? Does leapfrogging only happen between threads that feel like peers, not between a strong conviction and a stray thought?

Also worth sitting with: leapfrogging is *not* resolution. Neither ring wins — they just keep trading the lead indefinitely, in principle forever if undisturbed by viscosity. If this is what you're introspecting, it would mean some pairs of intuitions are structurally non-resolving — not "waiting for enough evidence to settle," but perpetually alternating dominance by nature. That's a stronger and stranger claim than "I have two conflicting intuitions I haven't resolved yet." Is that closer to what it actually feels like — an oscillation with no natural endpoint?

Reconnection, applied

This is the one I'd push on hardest because it's the most specific. Real vortex reconnection requires the two cores to actually come close enough that viscous effects let them briefly merge topologically — it's not "two ideas influencing each other," it's the field lines *breaking and rejoining in a new configuration*. The physically faithful prediction is: reconnection should produce a byproduct — Jung's tradition doesn't really have this in its vocabulary (contrast with, say, synthesis or dialectic, which imply resolution of two into one). A true reconnection event should leave behind smaller "off-cast" rings distinct from both parents — something you'd introspect as a minor, tangential insight that shows up

alongside the main resolution, not contained in it. Have you noticed that pattern specifically — a collision between two intuitions leaving a small "spare" insight neither one predicted on its own?

If you want to eventually operationalize this (tying back to your measurement interest): leapfrogging and reconnection give you distinguishable *signatures*, not just vibes. Leapfrogging predicts alternation without merger — you'd look for oscillating dominance in something measurable (which idea a person acts on/reports as primary, flipping back and forth). Reconnection predicts a discrete byproduct at a collision event — you'd look for a third, unplanned idea appearing right at the moment two threads meet, distinct from either. Those are actually different enough experimentally that you could in principle tell them apart in real data, human or model, which is more than most of this framework usually offers.

With ring vortexes, what are the cores referring to? I'm imagining smoke rings for most of this visually in my head, but usually those are donut shaped

Clarified vortex core as the region of most intense fluid circulation

Good question to nail down, since it changes what "reconnection" even means.

The core is the region of concentrated vorticity — not the visible smoke itself.

In a smoke ring, what you see is smoke trapped and carried along by the rotating fluid motion, but the actual fluid dynamics structure is a thin, roughly circular tube of rotating fluid — like a tiny cylinder bent into a circle and joined at the ends, forming the donut shape you're picturing. The "core" is that thin tube — specifically, the central axis running through the middle of the tube, where the rotational velocity (vorticity) is most concentrated. Velocity is highest near the core and falls off as you move away from it.

So picture the donut, but the actual physical structure is the thin ring-shaped tube that makes up the donut's "dough," not the hole and not the outer surface. The smoke you see is just tracer material caught in the flow — a visualization aid, not the vortex itself. A vortex ring would exist even with no smoke in it; the smoke just makes it visible.

Why this matters for "reconnection":

Reconnection specifically happens at the level of that thin rotating tube — when two rings get close enough, the tubes (not the smoke, not the outer donut boundary) can locally break and rejoin. So if you're mapping this onto Ni, the "core" would correspond to whatever the concentrated, structuring element of an intuition is — not the felt content or symbolic image (that's more like the smoke, the visible trace), but whatever is actually

doing the organizing work underneath it. That's worth pinning down for your model: what do you think the smoke is, in your framework, versus what's the actual rotating core?

Hmm, smoke is just tracer material for sure, but the core would be the expansion of the original pulse right?

Reconsidered vortex core formation mechanics and shear layer dynamics

Close, but the actual mechanism is more specific and probably more useful for your model.

It's not the pulse expanding — it's the edge of the pulse rolling up.

When a pulse of fluid gets pushed through an opening, the fluid at the *boundary* of that pulse moves against the still fluid around it, creating shear (a velocity difference at the edge). That shear causes the boundary layer to curl — like the edge of a rug catching and rolling as you push it forward. That curling edge is what consolidates into the core. It's not the whole pulse expanding outward into a ring; it's specifically the *skin* of the pulse folding in on itself around the entire perimeter simultaneously, forming the tube.

Meanwhile, the fluid that was in the *center* of the original pulse doesn't become the core at all — some of it gets entrained (pulled into) the rolling edge as it curls, and the rest just gets carried along inside the ring as a bubble of fluid the ring transports forward, distinct from the rotating core itself.

If you map this onto your model:

This suggests three components, not two — not just "core" and "smoke," but:

1. The *interior* of the original Se-pulse — content that gets carried along but isn't itself doing the structuring.
2. The *edge* of the pulse — the part that actually experiences "shear" against whatever it's moving through, and that shear is what curls into the core.
3. The *core* itself — the consolidated structure, made from that curled edge, which is what actually persists as "Ni" after the pulse is gone.

That reframes your question about what Ni's core is made of: it's not made from the "middle" of an experience, it's made from friction at the *boundary* of an experience against something else — context, contrast, whatever the surrounding "stationary fluid" represents for you. Does that change anything — is there a natural candidate in your model for what the "surrounding fluid" is, the thing the Se-pulse is shearing against to generate the roll-up?

The Se-pulse would be acting upon other Se-pulses, there would be no stationary fluid.

Reconsidered vortex dynamics without stationary reference frame

That actually still works physically — you don't need a stationary background, you need a *relative velocity difference*. Shear comes from two masses of fluid moving at different speeds or directions relative to each other, not from one being still. So if Se-pulses are acting on other Se-pulses, the roll-up would happen at the interface between any two pulses moving differently — the "stationary fluid" was never doing anything special except providing a reference frame with a velocity difference; other moving pulses can supply that difference just as well.

This actually changes the picture in an interesting way: it means core formation isn't a special event requiring a quiet backdrop — it's happening constantly, anywhere two Se-pulses meet with different speed/direction/intensity. Which fits better with the idea that Ni-structures are forming continuously in the background of ordinary experience, rather than only during rare, distinct "insight" moments.

It also raises a real question worth sitting with: if there's no stationary reference and everything is pulses interacting with other pulses, then a core's properties (tight/fast vs loose/slow, from a few messages back) would depend entirely on the *relative* difference between the colliding pulses, not on any pulse's own properties in isolation. That's actually a fairly strong, specific claim — it would mean an Ni-core's character says nothing about the "input" that formed it in isolation, only about the contrast between two inputs. Does that match your model, or does a single Se-pulse ever produce a core on its own, without needing another pulse to shear against?

I'll flag honestly: we're now several layers deep into extending a physical analogy past where the physics itself was ever meant to generalize, and I want to make sure I'm still being useful rather than just fluently extending metaphor for its own sake. This is a genuinely fun and generative exercise, and I don't think there's anything wrong with it — but if the goal is eventually measurement, it's worth periodically asking which of these structural claims you'd actually bet on being true versus which ones are just where the metaphor happens to lead.

It's mostly just confirming ideas I already hold, putting them into concrete language. A few of these are clarifying on what's happening. I'd bet on most being true, though it's separating Ni and Se even though they are so intimately related that makes it murky. The claim you say about Ni only saying something about the contrast between two inputs is interesting, a lot of the anecdotal descriptions of Ni speak of it as holding meaning, which is a nebulous term physically.

Reconciled Ni/Se entanglement while grounding nebulous meaning-holding concept

That tension you're naming — Ni and Se being too intimately related to cleanly separate — is worth taking seriously rather than smoothing over, because it might be telling you something about where the metaphor's boundary actually is, not just a modeling inconvenience.

On "holding meaning" as a physical claim:

Here's one honest way to ground it: in the vortex model, the core doesn't "contain" meaning as a stored thing — it *is* a stable pattern that persists across time and survives interaction with other structures (leapfrogging, reconnection) without dissolving back into undifferentiated flow. If you wanted a physically-grounded stand-in for "meaning" here, the closest fit isn't a substance held inside the ring, it's *invariance* — whatever stays constant about the core's identity across those interactions is what "meaning" would correspond to. That reframes "holding meaning" as "maintaining a recognizable identity under transformation," which is actually a real, well-studied concept in physics and math (this is closer to what "invariant" means in dynamical systems) — much less mystical than it sounds in the Jungian literature, and possibly testable: you'd look for some property of a person's articulated insight that stays stable even as the surrounding content/context shifts.

On the Ni/Se separation being murky:

This might not be a flaw in the model — vortex ring formation itself doesn't have a clean boundary between "core" and "the flow that formed it" either; the core is continuous with the fluid, just a region of concentrated rotation within it, not a separate object. If Ni is Ni is genuinely just a *regime* or *pattern* within Se-flow rather than a distinct substance, the murkiness you're describing might be the physically correct answer, not a gap in the analogy. That would actually be a stronger and more falsifiable claim than treating them as two separate interacting things: it would predict you should never find "pure Se" or "pure Ni" in isolation, only flow at different degrees of rotational concentration — a spectrum, not a duality. Does that match your sense of it, or does it feel like there really are moments of one without the other?

I think in the introspective experience, it can feel like they are separate and can be compartmentalized. But! There are other interactions between the other cognitive functions might smooth over that introspective experience, but I do think they are fundamentally linked

Gauged collaborative momentum and calibrated engagement depth

That distinction is actually important and worth keeping sharp: introspective *separability* and structural *separability* can come apart, and your fluid model gives you a clean reason why. A rotating core embedded in flow is still, locally, distinguishable — you can point to it, describe its motion, treat it as "a thing" — even though it's not ontologically separate from the fluid it's made of. So "feels separate, isn't separate" isn't a contradiction in the model, it's exactly what you'd expect from a regime-within-a-continuum rather than two distinct substances. The introspective compartmentalization might just be attention resolving at the scale of the core, not the whole flow field — you're perceptually zooming into the rotating region and treating its boundary as a real edge, the way you'd point to "a wave" in the ocean without the wave being separable from the water.

Your caveat about the other functions smoothing over the introspective picture is the more interesting complication, though — it suggests the two-body Se/Ni picture we've been using is a simplification of a much higher-dimensional flow (Te, Fe, Ti, Fi all contributing their own currents), and what you introspect as "Ni compartmentalized from Se" might really be "Ni compartmentalized from the *combined* flow of everything else," with Se just being the loudest or most proximate current. That's a testable difference in principle: if you ever tried to formalize this, you'd want to know whether apparent Ni/Se separation collapses once you account for the other functions' contributions, or holds up even then.

We've built a genuinely rich, internally consistent structural model here. If you want, next time we could try stress-testing it against a real example — take one specific remembered instance of an insight forming, and walk it through the model step by step (pulse, shear, roll-up, core, maybe a reconnection) — that's usually where a model's real gaps show up, more than continuing to extend the physics abstractly.

Can you give me a summary of the things we've gone through in this chat so I can take it into another chat? You can split it into current scoped plan (with readiness to go through "test 1") and Ni-Se metaphor/testable ideas as possible groups?

Partitioned discussion into two categorical frameworks

Summary for carrying into another chat

Group 1: Scoped near-term plan (mech interp path)

Background context: Cog sci degree (York), senior data platform engineer at Canadian Tire, applied to MATS last year (didn't get in), math is workable but not strong, went

through much of ARENA/Karpathy material last year.

Decision made: Prioritizing mechanistic interpretability over the alternative "AI + education research" path (an Anthropic Education Labs role was considered and set aside as a fallback/parallel option, not primary), because interp directly builds toward the long-term research question in Group 2.

Test 1 — ready to execute, not yet run (phone-only right now, needs a computer):

- **Question:** Is there an identifiable point during LLM generation where the model's internal trajectory becomes resistant to revision — i.e., does "momentum" build the deeper into a reasoning chain the model goes, such that a revision cue ("wait, let me reconsider") can only sometimes actually redirect it?
- **Sub-question 1 (binary, per-example):** Does a revision cue correspond to real internal change, or is it surface-only language?
- **Sub-question 2 (the actual momentum test):** Does correction/patching success rate drop off as the revision point occurs later in the trajectory?
- **Setup:**
 - `pip install transformer_lens torch`
 - `model = transformer_lens.HookedTransformer.from_pretrained("gpt2-small")`
(CPU-fine, ~500MB download)
 - Sanity check: `model.generate("The capital of France is", max_new_tokens=10)`
- **Method:**
 1. Build ~10 simple arithmetic/logic prompts where GPT-2 small errs, followed by a natural revision phrase (e.g. `"12 + 15 = 25. Wait, let me recompute. 12 + 15 ="`).
 2. Use `model.run_with_cache()` to get logits/activations at each token.
 3. Compare logit distribution for the answer token before vs. after the revision cue — did the predicted top token actually change?
 4. Vary how deep into the trajectory the revision cue appears; check whether correction success declines with depth.
- **Next step when picking this back up:** debug/build the code together step by step.

Group 2: Ni/Se vortex-ring metaphor (long-term, generative/interpretive — not yet operationalized)

Core structural model built so far:

- Ni modeled as a torus/vortex-ring structure; Se-pulses pass through and generate it.

- Vortex ring formation: an Se-pulse's *edge* (not its interior) experiences shear against other Se-pulses (no stationary reference needed — relative velocity difference between pulses is sufficient) and rolls up into the core. The pulse's interior is carried along but isn't the structuring element.
- Core = concentrated rotating structure (not the "smoke," which is just tracer/visible content); this is the candidate physical stand-in for what Ni "is" once formed.
- Propagation speed vs. core size trade-off → hypothesized as the key differentiator between primary-Ni and primary-Se cognitive styles.
- Viscous decay (rings dissipate without re-energizing) → attributed to Se's role.
- **Leapfrogging** (two similar-strength rings alternate lead, indefinitely, without merging) → flagged as one of two most interesting behaviors. Prediction: should only occur between "peer-weight" insights, not a strong conviction vs. a stray thought; implies some intuition-pairs are structurally non-resolving rather than pending resolution.
- **Reconnection** (cores briefly merge/break/reform, sometimes throwing off smaller derivative rings) → the other most interesting behavior. Prediction: should leave a distinct "byproduct" insight at the collision point, separate from either parent idea — not present in standard Jungian vocabulary (contrast with "synthesis").
- "Meaning" reframed non-mystically as *invariance* — whatever property of an insight stays stable/recognizable across interactions (leapfrogging, reconnection) could be the physical stand-in for "holding meaning."
- Open tension (unresolved, flagged as important): Ni/Se feel introspectively separable/compartmentalizable, but may not be structurally separate — possibly a regime-within-continuum (like a vortex core within fluid) rather than two distinct substances. User believes they're fundamentally linked, with other cognitive functions (Te, Fe, Ti, Fi) potentially smoothing over the introspective picture and making true separability hard to assess directly.

Explicitly flagged as unresolved/not yet tested:

- Whether Ni-core properties depend only on the *contrast* between two colliding Se-pulses vs. any single pulse in isolation.
- Whether apparent Ni/Se separation would collapse if the other cognitive functions' contributions were accounted for.
- **Suggested next step (not yet done):** stress-test the model against one specific remembered real insight, walking it step-by-step through pulse → shear → roll-up → core → (possibly) reconnection, to see where the model's real gaps show up.

Standing caveat carried over from this chat: this is a rich, internally consistent generative/interpretive model, useful for the user's own thinking — but it has not been

reframed into falsifiable, measurable claims yet, and doing so (per earlier discussion) would be a long-horizon (PhD-thesis-shaped) research program, not a near-term project.

